Fiscal Year:	FY 2016	**Task Last Updated:**	FY 12/01/2016
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	Combined Effects of Microgravity Analogue Culture and Incremental Oxygen Levels on Bacterial Pathogen Adaptive Responses		
Division Name:	Human Research, Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Ground	**Solicitation / Funding Source:**	2013 HERO NNJ13ZSA002N-Crew Health OMNIBUS
Start Date:	02/01/2015	**End Date:**	08/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Sato, Kevin	**Contact Phone:**	650-604-1104
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Flight Program:			
Flight Assignment:	NOTE: End date corrected to 8/31/2016 per PI and HRP Element information (Ed., 11/28/16) NOTE: End date changed to 8/31/2017 per NSSC information; original end date was 1/31/2016 (Ed., 7/25/16) NOTE: End date changed to 8/31/2016 per E. Connell/HRP SHFH; original end date was 1/31/2016 (Ed., 4/15/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Barrila, Jennifer Ph.D. (Arizona State University) Ott, Mark Ph.D. (NASA Johnson Space Center)		
Grant/Contract No.:	NNX15AD67G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE this project is jointly funded by NASA's Human Research Program and Space Biology for collaborative purposes. Spaceflight and/or spaceflight analogue-induced alterations in virulence and gene expression in the foodborne pathogen Salmonella enterica serovar Typhimurium and pathogenesis-related characteristics in both S. Typhimurium and the opportunistic pathogens Pseudomonas aeruginosa, Staphylococcus aureus, and Candida albicans have been reported by our team. In combination with alterations in crew immune function, these data suggest an increased risk of infectious disease during spaceflight missions. Previous research indicates that these alterations in microbial virulence and pathogenesis-related characteristics may be stimulated by possible secondary/indirect environmental effects during culture in microgravity, including fluid shear and decreased mass transfer of nutrients and oxygen. Our results evaluating gene expression and stress responses in the aforementioned pathogens cultured in spaceflight and/or the spaceflight-analogue, Rotating Wall Vessel (RWV), suggest that a limited transfer rate of oxygen may induce or exacerbate alterations in microbial virulence and pathogenesis-related characteristics during culture in these environments. The RWV bioreactor produces a low fluid shear, low turbulence environment for suspension culture (that we termed low shear modeled microgravity/LSMMG) that models aspects of spaceflight. We hypothesize that the combined effect of LSMMG culture and incremental oxygen levels will alter pathogenesis-related stress responses and gene expression of Salmonella and unveil novel connections between these environments in a way that neither approach alone or conventional culture could achieve. In this ground-based study, we propose to 1) Characterize the impact of LSMMG and incremental oxygen levels on targeted pathogenesis-related stress responses in Salmonella, including oxidative and acid stress, macrophage survival, and invasion, and 2) Characterize the effect of LSMMG and incremental oxygen levels on global gene expression profiles in response to these environments. This study addresses Integrated Research Plan (IRP) Gap AEH (Advanced Environmental Health)14: We need to determine how physical stimuli specific to the spaceflight environment, such as microgravity, induce unique changes in the dose-response profiles of expected medically significant microorganisms.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	We anticipate that findings from the proposed studies may be extended to other microorganisms, thereby advancing our knowledge of 1) important principles in pathogenesis that can be translated toward novel strategies to mitigate infectious disease risk to the crew, and 2) the role of the microenvironment on cellular responses from both a molecular and ecological perspective.
Task Progress:	NOVEMBER 2016--SUPPLEMENTAL FINAL REPORTING • We observed that under microaerobic culture conditions in the RWV, a substantial amount of gas production by bacteria accumulates in the reactors (as evidenced by significant bubble formation) during both stationary and late log phase. We reasoned that this effect would be further exacerbated during anaerobic culture. However, this effect is not observed under aerobic cultures of the same strain for either growth phase. • During this work, we extended our previous findings under aerobic conditions to include growth phase regulated fluid shear induced responses of Salmonella Typhimurium to additional pathogenesis-related stressors during RWV culture. For these stressors, we observed distinct phenotypic differences in Salmonella stress survival between LSMMG and control conditions that suggest an important role in pre-adaptation of pathogenesis-related phenotypes in the infected host. ANNUAL REPORTING AS OF AUGUST 2016 The start of the project was originally delayed several months due to release of funding, but project funding has now been received and project deliverables are well underway. • We have completed growth profiles at log and stationary phase under aerobic and microaerophilic conditions to ensure that LSMMG and control RWV Salmonella cultures are in the same phase of growth for subsequent end point analysis. • Stress tests and cell culture infections were done with microaerophilic LSMMG and control cultures, but technical challenges with stationary phase cultures grown under reduced oxygen were complicated by the production of gas, which makes data interpretation challenging. We thus focused our subsequent studies on log phase cultures under both aerobic and microaerophilic conditions. • We have observed phenotypic differences in Salmonella stress survival as function of oxygen in log phase cultures. • Data analysis is ongoing. We are still finalizing our studies, thus no data has yet been published.
Bibliography Type:	Description: (Last Updated: 07/02/2025)